

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (E&C) (Credit System) Degree Examinations
January 2023

20EC505 - OPERATIONS RESEARCH

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Define Operations Research. List and explain the different phases of Operations Research. 12 L*2 *1 1
- b) A company produces two products A and B, and possess raw materials 400 quintals and 450 labour hours. It is known that 1 unit of product A requires 5 quintals of raw materials and 10 man hours and yields a profit of Rs. 45. Product B requires 20 quintals of raw materials, 15 man hours and yields a profit of Rs.80. Formulate the LPP. 8 L4 1 2
2. a) Solve the following LPP using Graphical method.
Maximize subject to $Z = 80x_1 + 55x_2$
 $4x_1 + 2x_2 \leq 40$
 $2x_1 + 4x_2 \leq 32$
 $x_1 \geq 0, x_2 \geq 0$ 12 L3 2 1
- b) What are the different applications of Operations Research? 8 L2 1 1
3. a) A manufacturer of furniture makes two products-chairs and tables. Processing of this product is done on two machines A and B. A chair requires 2 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours of time per day available on machine A and 30 hours on machine B. Profit gained by the manufacturer from a chair and a table is Rs.2 and Rs.10 respectively. What should be the daily production of each of two products? Solve using graphical method. 12 L4 2 2
- b) Explain the following: i) Feasible solution ii) Feasible region
iii) Optimal solution iv) unbounded solution 8 L2 2 1

Unit – II

4. a) Solve the following LPP by simplex method
Maximize $Z = 80x_1 + 55x_2$
subject to $4x_1 + 2x_2 \leq 40$
 $2x_1 + 4x_2 \leq 32$
 $x_1 \geq 0, x_2 \geq 0$ 10 L3 3 1
- b) Solve the following LPP by Big-M Method.
Maximize $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \geq 2$
 $x_1 + 3x_2 \leq 3$
 $x_2 \leq 4$
 $x_1 \geq 0, x_2 \geq 0$ 10 L3 3 1
5. a) Determine the initial basic feasible solution to the following transportation problem using i) North-west corner Rule ii) Vogel's Approximation Method.

Warehouse→ Factory↓	W ₁	W ₂	W ₃	W ₄	Factory capacity
F1	19	30	50	10	7
F2	70	30	40	60	9
F3	40	8	70	20	18
Warehouse Requirement	5	8	7	14	

- b) Explain the steps involved in Two-Phase Method. 12 L3 4 1
8 L4 3 1

P.T.O.

6. a) Solve the following LPP using Two-Phase Method.

$$\text{Maximize } Z = 5x_1 + 8x_2$$

$$\text{subject to } 3x_1 + 2x_2 \geq 3$$

$$x_1 + 4x_2 \geq 4$$

$$x_1 + x_2 \leq 5$$

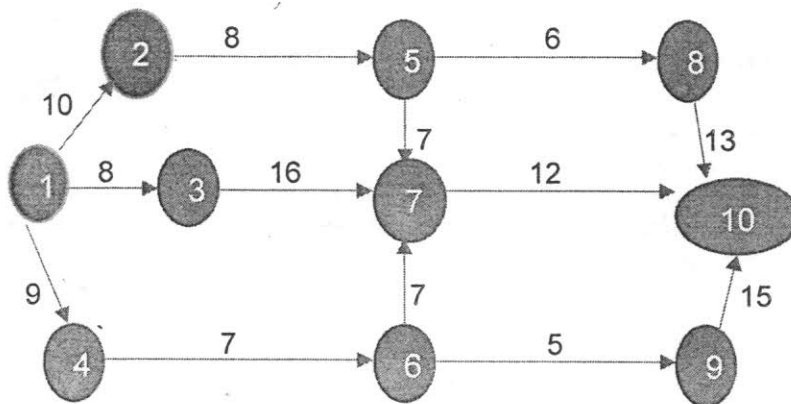
$$x_1 \geq 0, x_2 \geq 0$$

- b) Write a note on degeneracy in transportation problem.

12	L3	3	1
8	L2	4	1

Unit – III

7. a) Determine the early start and late start in respect of all node points and identify critical path for the following network.



10	L3	5	1
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- b) A car hire company has one car at each of five depots a, b, c, d and e. A customer requires a car in each town namely A, B, C, D and E. The distance (km) between depots (origin) and towns (destination) are given in the following distance matrix.

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	55	35	70	80	105

Solve the above assignment problem so as to minimize the distance.

10	L3	5	1
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8. a) Explain the steps involved in finding the critical path using CPM.

8	L2	5	1
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- b) Solve the following instance of Assignment Problem.

	R1	R2	R3	R4
C1	9	14	19	15
C2	7	17	20	19
C3	9	18	21	18
C4	10	12	18	19
C5	10	15	21	16

12	L3	5	1
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
